Code No: 138FY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July/August - 2022
ELECTRONIC MEASURING INSTRUMENTS
(Common to CE, EEE, ME, CSE)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Define the terms repeatability and reproducibility with respect to measuring instruments.
b) Describe the static characteristic of a measurement instrument. [7+8]

2.a) What are the different type of the errors in the field of instrumentation and measurements?
b) The expected value of the voltage across a resistor is 80 V. However, the measurement gives a value of 79 V. Calculate (i) absolute error, (ii) % error, (iii) relative accuracy, and (iv) % of accuracy. [9+6]

3.a) What is a signal generator? Explain its working with neat diagram.
b) Compare the signal generator and function generator. [10+5]

4.a) Draw the simple diagram of a pulse generator and explain it briefly.
b) Describe the RF generator with diagrams? What are its specifications? [8+7]

5.a) Describe the construction details and working principle of D'Arsonval type of instrument. What are its advantages and disadvantages?
b) How to extend the range of ohm meter? Explain any one method. [8+7]

6.a) Draw the basic block diagram of an oscilloscope and explain the functions of each block.
b) In a CRT, the distance between the plates is 1 cm, the length of the deflecting plates is 4.5 cm and the distance of the screen from the centre of the plates is 33 cm. If the accelerating voltage is 300 V and deflecting voltage is 50 V, find the following:
i) Velocity of electron reaching the field
ii) Deflection produced on the screen
iii) Deflection sensitivity. [9+6]

7.a) Describe the pen-driving mechanism with diagram.
b) What is the *X*-*Y* recorder? How it is differ from strip chart recorder. Describe it advantages and application. [8+7]

8.a) What are the points to be considered for selecting transducer?
b) Describe the construction and working of LVDT. [6+9]

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